reMIND

powered by Remi

meet the team

Spencer Will

[will2sa@mail.uc.edu]

5th year Computer Science student

Hannah Laman

[lamanhm@mail.uc.edu]

5th year Computer Science student

Dylan Francis

[francidb@mail.uc.edu]

5th year Computer Science student

Austin Spencer

[spencar@mail.uc.edu]

5th year Computer Science student

PROJECT ADVISOR: Prof. Badri Vellambi [vellambn@ucmail.uc.edu]

project overview

reMIND is a wearable AI companion for people with dementia that provides personalized memory support, reminders, and guidance, while offering emergency assistance and caregiver connectivity. It uses a friendly, customizable persona (Remi) to help users recall who they are, where they are, and what they're doing, adapting to their cognitive needs over time.

project purpose + goal statements

project purpose:

to design and develop a wearable AI companion that assists individuals with dementia by providing personalized memory support, orientation guidance, and emergency communication. the system aims to enhance user independence, safety, and emotional well-being through accessible voice interaction and seamless caregiver connectivity.

01

develop a natural voice-based AI interface to assist users with memory recall and orientation.

02

build a secure and adaptive memory system that protects sensitive user data.

03

create a connected caregiver app enabling real-time alerts, monitoring, and emergency support.

project abstract

reMIND, powered by Remi, is an assistive technology designed to enhance memory support, safety, and communication for individuals with dementia. This voice-activated device allows users to recall personal information, receive reminders, and access comforting conversational interactions in a customizable voice. Connected to a caregiver mobile app, Remi securely manages data synchronization, emergency alerts, and real-time monitoring. Emphasizing ethical design, privacy compliance (i.e.., HIPAA), and affordability, the project integrates AI-driven speech recognition, text-to-speech, and cloud-based memory retrieval. Together, these features aim to promote independence, dignity, and quality of life for dementia patients and their caregivers through intelligent, human-centered design.

intellectual merits

AI-enabled wearable system integrating real-time memory assistance, contextual awareness, and caregiver connectivity in one unified platform.

dual-user architecture designed specifically for both dementia patients and caregivers, addressing cognitive support and remote monitoring simultaneously.

context-aware intelligence leveraging time, location, and behavioral pattern analysis to deliver adaptive reminders and personalized prompts.

edge + cloud hybrid design optimizing responsiveness, scalability, and privacy in a sensitive healthcare setting.

interdisciplinary contribution bridging computer science, machine learning, UX design, and healthcare technology to address a growing societal need.

broader impacts

supports aging-in-place, enabling individuals with dementia to maintain independence longer and delay institutional care.

reduces caregiver burden through remote monitoring, behavioral insights, and proactive alerts.

improves safety outcomes via context-aware reminders and real-time location-based assistance.

promotes ethical, privacy-conscious AI in healthcare through secure data handling and explainable system design.

scalable framework for future AI-assisted health wearables addressing broader cognitive and aging challenges.

user stories

user	story
person living with dementia	as a person living with dementia, I want to ask the AI questions about who I am and what I'm doing, so that I can feel reassured and maintain independence.
caregiver	as a caregiver, I want to receive alerts and notifications from the AI system, so that I can monitor the safety and well-being of my loved one.
primary care provider	as a primary care provider, I want the AI companion to send me emergency alerts when the patient is confused or unsafe, so that I can respond quickly and ensure proper medical care.
family member	as a family member, I want the AI's voice to be customized to sound like someone familiar, so that my loved one feels comforted and less anxious.

project constraints

01 economic

limited funding restricts us to affordable hardware and open-source software, influencing design choices and scalability of the final product.

02 legal / security

privacy laws like HIPAA and data protection standards require secure storage, encrypted communication, and strict access controls to safeguard sensitive user information.

03 social / cultural

the device must be easy to use for people with dementia and adaptable to different languages, voices, and cultural preferences to ensure comfort and inclusivity.

project design

system overview:

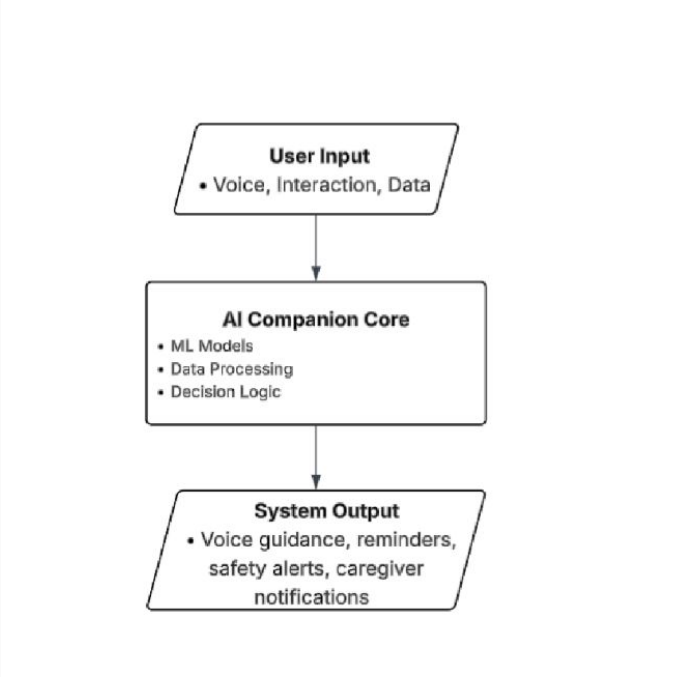
- Three-tier architecture:
  - watchOS patient app → REST/WebSocket API → Azure cloud services
- Real-time voice interaction via Azure Voice Live API
- HealthKit + location data streamed to backend
- Secure SQL + NoSQL storage for behavioral & reminder data
- Caregiver web frontend connected through system API

---

project implementation:

- watchOS app built in Swift + Xcode, tested on simulator
- REST + WebSocket endpoints implemented in remind-api
- Azure AI Foundry + SQL/NoSQL deployed via infra
- Caregiver frontend scaffolded with two-way reminder system
- Modular repo structure enabling scalable cloud + device integration

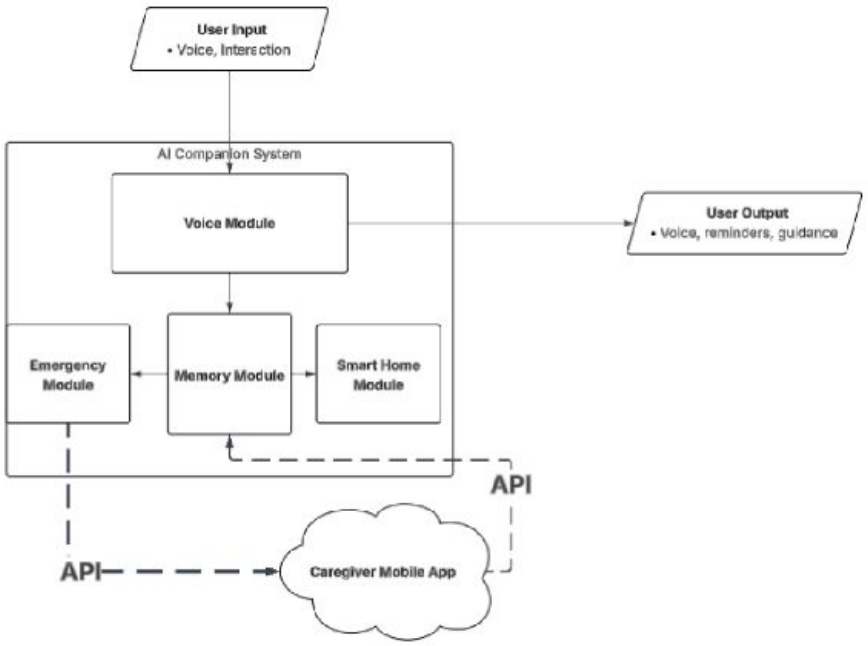
design diagrams: d0



Design D0: High-Level Overview

The system collects inputs (voice, sensors, caregiver app, smart home), processes them with AI and decision logic, and provides outputs (voice feedback, reminders, emergency alerts, caregiver updates).

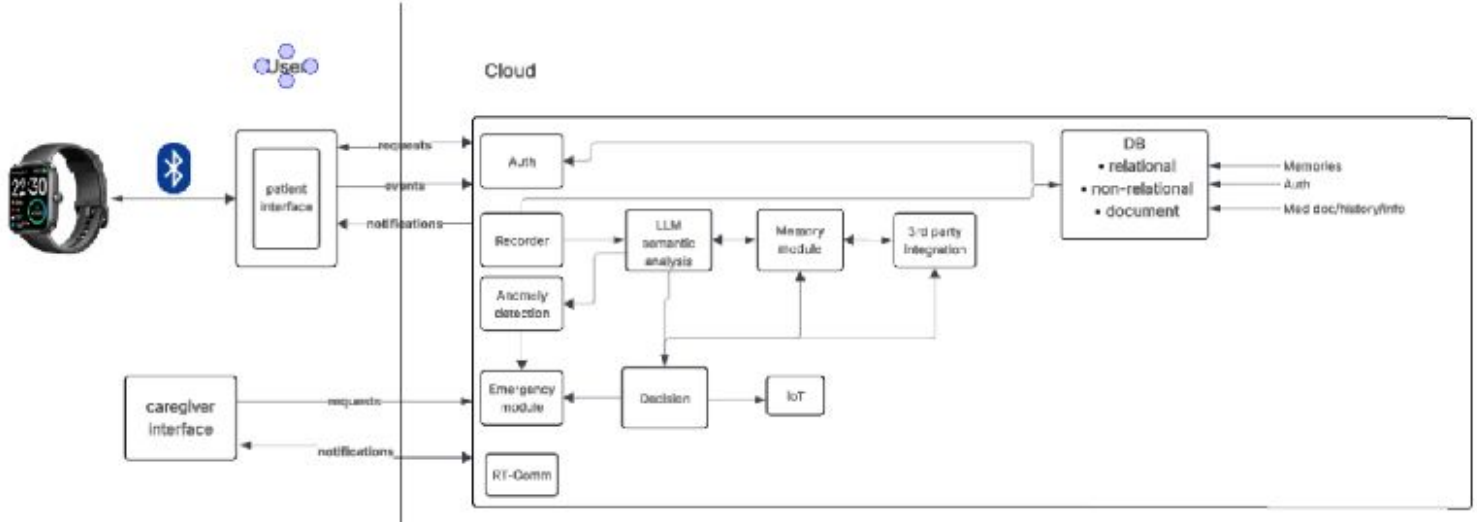
design diagrams: d1



Design D1: Subsystem Modules

This level introduces the major modules of the AI Companion system: voice interaction, memory support, emergency handling, caregiver app, and smart home integration.

design diagrams: d2



Design D2:

Design D2 provides the most detailed view of the Wearable AI Companion system, expanding each module into its subcomponents and showing how data flows between them. This level highlights the full integration of all subsystems, emphasizing real-time support, personalization, and emergency responsiveness.

technologies

watchOS (Swift, Xcode)
- native Apple Watch application integrating HealthKit, location services, and voice interaction.

Azure AI Foundry + Voice Live API
- real-time speech-to-speech LLM interaction for contextual memory assistance.

Backend API (REST + WebSocket)
- real-time communication layer enabling live updates, reminders, and caregiver synchronization.

Azure SQL + NoSQL Databases
- structured reminder storage + scalable behavioral data management.

Cloud Infrastructure (Azure)
- deployed and managed via modular infrastructure configuration for scalability and security.

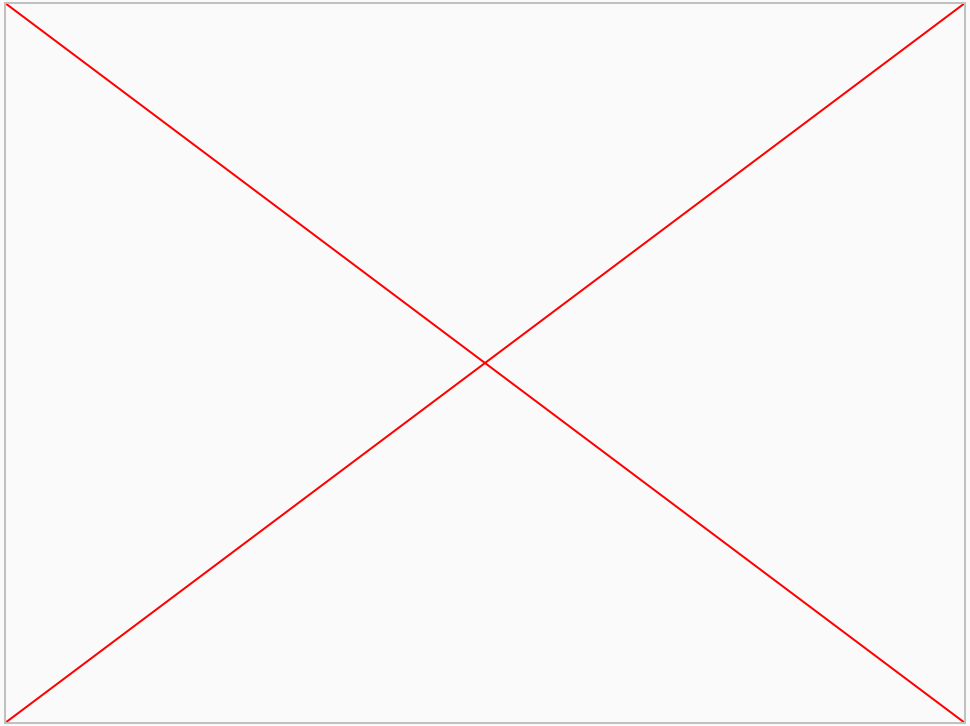
Web Frontend (Caregiver App)
- two-way reminder system with live health and location updates.

project progress + milestones

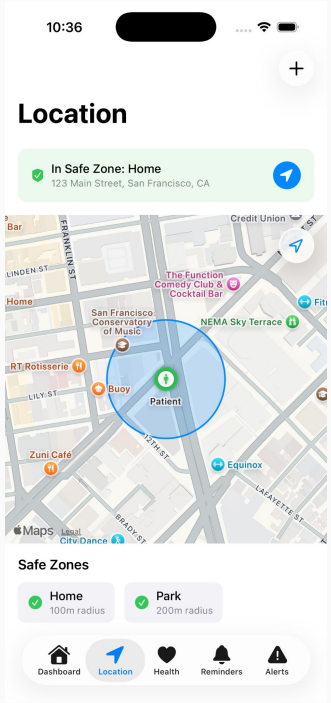
<u>what we have accomplished so far:</u>

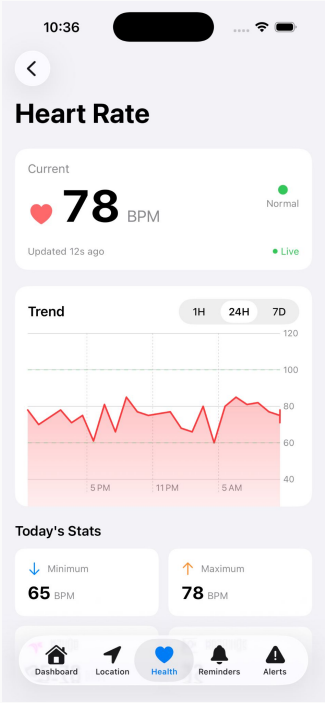
Deployment of Azure Resources	Patient watchOS app Created and Designed	Caregiver Frontend Design and Scaffolded	System API Created
Azure AI Foundry project	Integration with Azure Voice Live API	Healthkit updates	REST endpoint definitions
Azure SQL Database & NoSQL Database setup	Live speech-to-speech LLM interaction	Location updates to watch	WebSocket endpoint definitions
	User settings & localization support	Two-way reminder system	
	Build & test on xCode watchOS simulator		

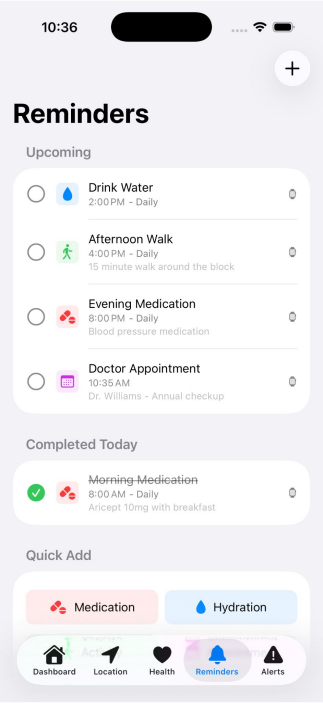
project demo

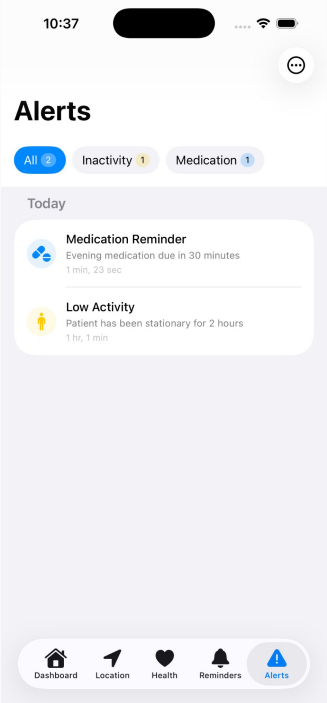


caregiver app progress









project progress

what we will accomplish before EXPO:

Feb 25

- context + backend data integration to watch app
- verified watch ↔ API communication

March 6

- location tracking implemented on watch
- caregiver app displays live location updates

March 18

- two-way reminder system fully functional
- watch acknowledgment synced to backend

March 27

- health data syncing watch → backend → caregiver
- full end-to-end system testing complete

project results

results achieved so far:

- watchOS patient app designed and functional in simulator
- Azure AI Foundry + Voice Live API integrated for real-time speech interaction
- Backend REST + WebSocket API implemented
- Azure cloud infrastructure deployed (AI, database, API services)
- Caregiver frontend scaffolded with reminder framework

remaining work to final demo:

- **March 6**: location tracking integrated (watch → caregiver)
- **March 18**: two-way reminder system fully operational
- **March 27**: health data syncing complete + full system test
- **April 3**: final demo build complete

project challenges

cloud resource & credit limitations

- Azure AI services required active cloud deployment
- student credit limits constrained long-term testing and scaling
- real-time voice + AI services increased infrastructure dependency

our contribution / resolution:

- carefully monitored usage during development
- prioritized core feature implementation over non-essential expansion
- structured architecture to remain scalable beyond student-credit environment

building trust in an AI companion

- AI-generated responses must feel supportive, not confusing
- Risk of hallucinations or inappropriate suggestions in a healthcare-adjacent setting
- Need for predictable, safe interactions

our contribution / solution:

- constrained prompt structure for AI responses
- focused on task-based assistance (reminders, context) rather than open-ended advice
- designed system to route structured data instead of relying solely on generative output

THANK YOU!

BRANDING GUIDELINES

Fonts:

- Inter
- Code Pro (logo)

#a8dcab

reMIND
powered by Remi

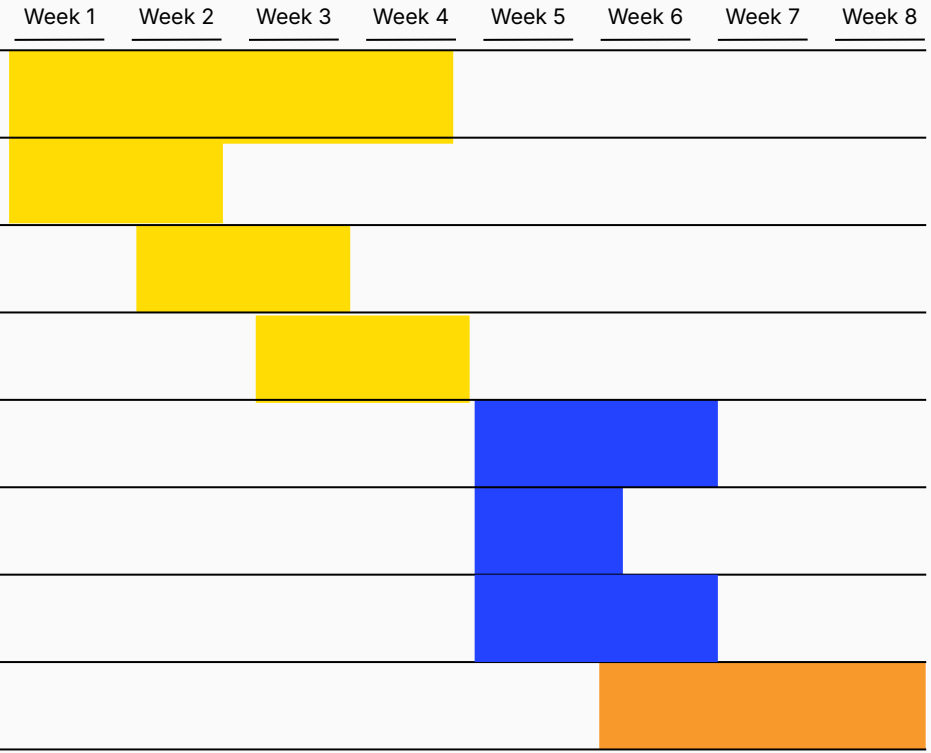
reMIND
powered by Remi

reMIND
powered by Remi

reMIND
powered by Remi

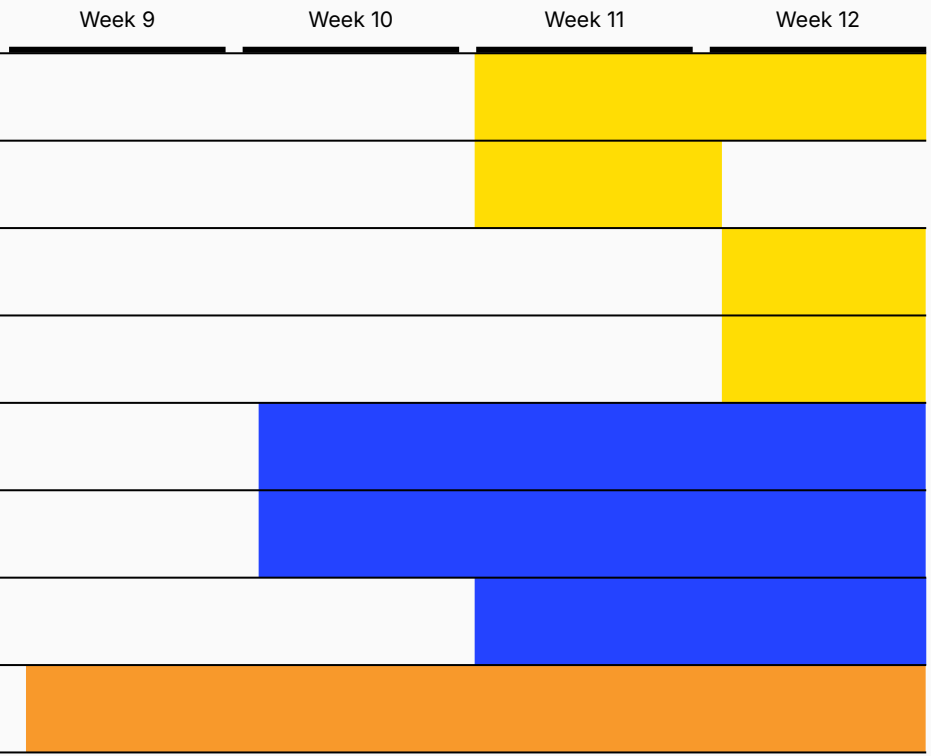
project progress - current state

	Duration	Start	End
System Planning and Architecture	4 Weeks	Week 1	Week 4
Requirements / Team Setup	2 Weeks	Week 1	Week 2
Design Diagrams	2 Weeks	Week 2	Week 3
Task List + Effort Matrix	2 Weeks	Week 3	Week 4
Constraints / Design Preparation	2 Weeks	Week 5	Week 7
Project Constraints Essay	1 Week	Week 5	Week 6
Milestones & Timeline	2 Weeks	Week 5	Week 7
Presentation Prep	2 Weeks	Week 6	Week 8



project progress - end of term

	Duration	Start	End
Final Advisor Coordination	2 Weeks	Week 11	Week 12
Advisor Meeting Preparation	1 Week	Week 10	Week 11
Advisor Assessment Disc	1 Week	Week 11	Week 12
Submission of Advisor Assessment	1 Week	Week 11	Week 12
Repository and Documentation Wrap-Up	3 Weeks	Week 10	Week 12
Organize Repository	3 Weeks	Week 10	Week 12
Archive Fall Deliverables	2 Weeks	Week 11	Week 12
Spring Preparation	4 Weeks	Week 9	Week 12



division of work

Spencer Will

Voice Interaction

Implement and test the voice recognition and text-to-speech module.

Hannah Laman

Caregiver App

Design caregiver app UI and establish communication with wearable device.

Dylan Francis

Memory Support

Build database structure for storing and recalling memory data.

Austin Spencer

Emergency Handling

Prototype and test emergency detection and alert protocols.